GRADE 9 · MATHEMATICS ENGLISH MEDIUM

MARKING SCHEME

Model Paper — 02

Third Term Test

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Note (for teachers):

- Part I – A: 2 marks per question (normally 1 mark for the correct method and 1 mark for the correct final answer). Total 30 marks.
- Part I – B: 10 marks per question. Total 20 marks. Part II: 10 marks per question; candidates answer any **five**. Total 50 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. For measured values (constructions, scale drawings) accept the ranges given.

PART I – A — Answers

1. $1101_2 = 13$, $111_2 = 7$, $13 + 7 = 20 = 10100_2$ (or by column addition in base two: $1101 + 0111$)

Final answer: 10100_2

(2 marks)

2. $(2^3)^2 = 2^6$; $2^6 \times 2^{-4} = 2^{6-4} = 2^2$

Final answer: 4

(2 marks)

3. Third decimal digit is 7 (≥ 5), so round the second decimal digit 6 up to 7

Final answer: 3.47

(2 marks)

4. $T_{20} = 4 \times 20 - 1 = 80 - 1$

Final answer: 79

(2 marks)

5. $\frac{3}{4} \times N = 27$; $N = 27 \times \frac{4}{3}$

Final answer: 36

(2 marks)

6. 18% = Rs 54, so 1% = Rs 3 ; 100% = 3 × 100

Final answer: Rs 300

(2 marks)

7. $x^2 - 3x + 5x - 15$

Final answer: $x^2 + 2x - 15$

(2 marks)

8. Volume = $2 \times 1.5 \times 1 = 3\ m^3$; $1\ m^3 = 1000$ litres

Final answer: 3000 litres

(2 marks)

9. The points equidistant from two fixed points lie on the line that bisects AB at right angles

Final answer: the perpendicular bisector of AB

(2 marks)

10. $s = \frac{5 + 7 + 8}{2} = \frac{20}{2}$

Final answer: s = 10

(2 marks)

11. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$; the square of the longest side equals the sum of the squares of the other two sides (converse of Pythagoras)

Final answer: right-angled triangle (hypotenuse 25 cm)

(2 marks)

12. $3.2 \times 10^5 = 320\ 000$ g ; $320\ 000 \div 1000 = 320$ kg

Final answer: 320 000 g = 320 kg

(2 marks)

13. Sum of exterior angles = 360° ; each exterior angle = 360° ÷ 8

Final answer: 45°

(2 marks)

14. Closed (filled) circle at −2 means $x \geq -2$; open circle at 3 means $x < 3$

Final answer: $-2 \leq x < 3$; integers: −2, −1, 0, 1, 2

(2 marks)

15. $\frac{2}{x} + \frac{3}{2x} = \frac{4}{2x} + \frac{3}{2x}$

Final answer: $\frac{7}{2x}$

(2 marks)

PART I – B — Answers and mark allocation

Question 1

(i) Rough sketch	
H, A and B shown with HA = 40 km on a bearing of 060° (north line at H, 60° marked) and AB = 30 km on a bearing of 150° (north line at A, 150° marked)	**1**
Points and distances correctly labelled; B drawn south-east of A so that HÂB looks like a right angle	**1**
(ii) Show that HÂB = 90°	
North lines at H and A are parallel. Angle between AH and the line AS (south) = 60° (alternate angles to the 060° bearing at H)	**1**
Angle between AS and AB = 180° − 150° = 30° ; HÂB = 60° + 30° = 90°	**1**
(iii) Distance HB	
$HB^2 = HA^2 + AB^2 = 40^2 + 30^2 = 1600 + 900 = 2500$	**1**
HB = 50 km	**1**
(iv) Scale diagram (1 cm = 10 km)	
HA = 4 cm drawn at 060° from the north line through H	**1**
AB = 3 cm drawn at 150° from the north line through A (perpendicular to HA), and HB joined (HB ≈ 5 cm; 4.9 – 5.1 cm accepted)	**1**
(v) Bearing of B from H	
North line at H drawn and the angle from north to HB measured clockwise with a protractor	**1**
Bearing ≈ 097° (exact value: $60° + \tan^{-1}(30 \div 40) = 60° + 36.9° = 096.9°$; 095° – 099° accepted)	**1**

Total marks: 10

Question 2

(i) Volume	
V = 1.2 × 0.8 × 1	**1**
= 0.96 m^3	**1**
(ii) Capacity	
1 m^3 = 1000 litres ; 0.96 × 1000	**1**
= 960 litres	**1**
(iii) Time to fill	
Time = 960 ÷ 40	**1**
= 24 minutes	**1**
(iv) Number of days	
960 ÷ 240	**1**
= 4 days	**1**
(v) Depth of water	
480 litres = 0.48 m^3 ; base area = 1.2 × 0.8 = 0.96 m^2 ; depth = 0.48 ÷ 0.96	**1**
= 0.5 m (50 cm) [check: 480 is half of 960, so the depth is half of 1 m]	**1**

Total marks: 10

PART II — Answers and mark allocation

Candidates answer any five questions. (10 marks each)

Question 1

(a)(i) Fraction spent on rent	
Remainder after food = $1 - \frac{2}{5} = \frac{3}{5}$	**1**
Rent = $\frac{1}{4} \times \frac{3}{5} = \frac{3}{20}$	**1**
(a)(ii) Fraction saved	
Spent = $\frac{8}{20} + \frac{3}{20} = \frac{11}{20}$	**1**
Saved = $1 - \frac{11}{20} = \frac{9}{20}$	**1**
(a)(iii) Salary and food	
$\frac{9}{20}$ of salary = Rs 27 000	**1**
Salary = 27 000 × $\frac{20}{9}$ = Rs 60 000	**1**
Food = $\frac{2}{5}$ × 60 000 = Rs 24 000 [check: 24 000 + 9 000 (rent) + 27 000 = 60 000]	**1**
(b) Commission	
Commission = $\frac{3}{100}$ × 2 500 000	**2**
= Rs 75 000	**1**

Total marks: 10

Question 2

(i) Missing values	
$y = 3 - x$: $x = 0 \rightarrow y = 3$; $x = 2 \rightarrow y = 1$	**1**
$y = x + 1$: $x = 1 \rightarrow y = 2$; $x = 3 \rightarrow y = 4$	**1**
(ii) Graphs (same coordinate plane)	
Both axes drawn and labelled with a suitable uniform scale (x from −1 to 4, y from −1 to 5)	**1**
Points of $y = 3 - x$ plotted correctly: (−1, 4), (0, 3), (1, 2), (2, 1), (3, 0), (4, −1)	**1**
Points of $y = x + 1$ plotted correctly: (−1, 0), (0, 1), (1, 2), (2, 3), (3, 4), (4, 5)	**1**
Both graphs drawn as straight lines through the points ($y = 3 - x$ slopes down to the right, $y = x + 1$ slopes up to the right) and labelled	**1**
(iii) Gradients	
Gradient of $y = 3 - x$ is −1	**1**
Gradient of $y = x + 1$ is 1	**1**
(iv) Point of intersection	
The lines cross at the common point (1, 2) of both tables (algebraically: $3 - x = x + 1 \Rightarrow x = 1$)	**1**
Coordinates (1, 2)	**1**

Total marks: 10

Question 3

(a)(i) $x^2 - 9x + 20$	
Two numbers with product 20 and sum −9: −4 and −5	**1**
$(x - 4)(x - 5)$	**1**
(a)(ii) $9a^2 - 16b^2$	
$(3a)^2 - (4b)^2$	**1**
$(3a - 4b)(3a + 4b)$	**1**
(a)(iii) $3x^2 + 6x$	
$3x(x + 2)$	**1**
(b) Simplify	
$\frac{5(x + 1) - 2(x - 3)}{10}$ (common denominator 10)	**1**
$\frac{5x + 5 - 2x + 6}{10}$	**1**
$\frac{3x + 11}{10}$	**1**
(c) $103^2 - 97^2$	
$(103 - 97)(103 + 97)$	**1**
$= 6 \times 200 = 1200$	**1**

Total marks: 10

Question 4

(a) 5(x − 2) − 3(x + 1) = 7	
5x − 10 − 3x − 3 = 7	**1**
2x − 13 = 7 ⇒ 2x = 20	**1**
x = 10 [check: 5(8) − 3(11) = 40 − 33 = 7]	**1**
(b) Simultaneous equations	
x + y = 25	**1**
x − y = 7	**1**
Adding: 2x = 32 ⇒ x = 16	**1**
y = 25 − 16 = 9 ; the numbers are 16 and 9	**1**
(c) Rectangle	
Breadth b cm, length (b + 4) cm ; 2(b + b + 4) = 36 ⇒ 2b + 4 = 18	**1**
b = 7 cm, length = 11 cm	**1**
Area = 11 × 7 = 77 cm^2	**1**

Total marks: 10

Question 5

(i) AB = 7 cm	
Line segment AB drawn with length 7 cm (6.9 – 7.1 cm accepted)	**1**
(ii) 60° at A	
Arc from A cutting AB; same radius arc from the cut point cutting the first arc (construction arcs shown)	**1**
Ray AX drawn through the arc intersection (angle BAX = 60°, accept ± 1°)	**1**
(iii) 45° at B, point C	
90° constructed at B (perpendicular to AB, on the same side as AX) with construction arcs	**1**
That 90° angle bisected with arcs to give 45° (ray BY)	**1**
Rays AX and BY meet; the meeting point is labelled C	**1**
(iv) Measurements	
$A\hat{C}B = 75°$ (180° − 60° − 45° ; 73° – 77° accepted)	**1**
AC ≈ 5.1 cm (sine rule: AC = 7 sin 45° ÷ sin 75° = 5.12 cm ; 5.0 – 5.2 cm accepted). For checking: BC ≈ 6.3 cm	**1**
(v) Bisector of $A\hat{C}B$	
Arc centred C cutting CA and CB; two equal arcs from those points intersecting inside the angle	**1**
Line drawn from C through the intersection point; each half-angle ≈ 37.5° (36.5° – 38.5° accepted)	**1**

Total marks: 10

Question 6

(i) Area of rectangle	
Area = 14×10	**1**
$= 140\ \text{cm}^2$	**1**
(ii) Area of semicircle (diameter DC = 14 cm, radius 7 cm)	
Area = $\frac{1}{2} \times \frac{22}{7} \times 7 \times 7$	**1**
$= 77\ \text{cm}^2$	**1**
(iii) Total area	
$140 + 77 = 217\ \text{cm}^2$	**1**
(iv) Perimeter (the side DC is not part of the boundary)	
Arc length = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm	**1**
AB + BC + AD = 14 + 10 + 10 = 34 cm	**1**
Perimeter = 34 + 22 = 56 cm	**1**
(v) Square with perimeter 56 cm	
Side = 56 ÷ 4 = 14 cm	**1**
Area = $14 \times 14 = 196\ \text{cm}^2$	**1**

Total marks: 10

Question 7

(a) Probabilities (8 equally likely outcomes: 1, 2, 3, 4, 5, 6, 7, 8)	
(i) Even numbers: 2, 4, 6, 8 $\Rightarrow$ P = $\frac{4}{8} = \frac{1}{2}$	**1**
(ii) Prime numbers: 2, 3, 5, 7 (1 is not prime)	**1**
P = $\frac{4}{8} = \frac{1}{2}$	**1**
(iii) Multiples of 3: 3, 6 $\Rightarrow$ P = $\frac{2}{8} = \frac{1}{4}$	**1**
(b) Marks: 45, 52, 60, 38, 52, 70, 65, 52, 41, 55	
(i) Mode = 52 (occurs three times)	**1**
(ii) Ascending order: 38, 41, 45, 52, 52, 52, 55, 60, 65, 70	**1**
Median = average of 5th and 6th values = (52 + 52) ÷ 2 = 52	**1**
(iii) Sum = 530	**1**
Mean = 530 ÷ 10 = 53	**1**
(iv) Range = 70 − 38 = 32	**1**

Total marks: 10